Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 6 — Trigonometric ratios
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第6章 · 三角比
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Triangle ABC has $AB = 9$ cm, $AC = 7$ cm and $\angle BAC = 60°$.

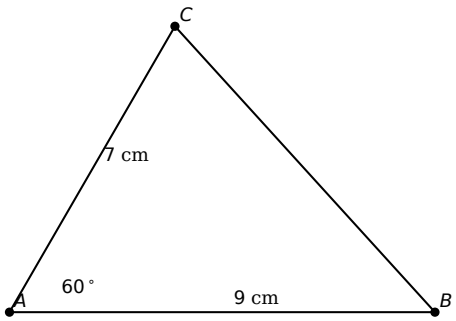


Figure 1

(a) Find the exact length of BC. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 1 is 5 marks)

2. P1-C06-R02 · P1 · Chapter 6

Triangle PQR has $PQ = 12$ m, $\angle PQR = 41^\circ$ and $\angle PRQ = 67^\circ$.

(a) Find $\angle QPR$. (2)

(b) Giving lengths to 3 significant figures, find

(i) PR, (2)

(ii) the perimeter of the triangle. (3)

(Total for Question 2 is 7 marks)

3.

In triangle ABC, $AB = 10$ cm, $BC = 8$ cm and $\angle BAC = 35°$.

(a) Use the sine rule to show that two values of $\angle ACB$ are possible. Give both values to 1 decimal place. (3)

(b) For the triangle in which $\angle ACB$ is obtuse, find AC to 3 significant figures. (3)

(Total for Question 3 is 6 marks)

4.

A triangular panel has sides 6 cm, 9 cm and 11 cm.

(a) Find its largest angle to 1 decimal place. (3)

(b) Find its area to 3 significant figures. (2)

(Total for Question 4 is 5 marks)

5.

Two paths leave a station S. A point A lies 120 m due east of S, and a point B lies 150 m from S on a bearing of 040°.

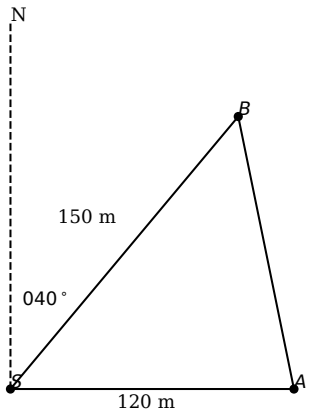


Figure 5

(a) Find $\angle ASB$. (2)

(b) Find AB and the area of triangle SAB, both to 3 significant figures. (4)

(Total for Question 5 is 6 marks)

A vertical tower stands on horizontal ground. From point A, the angle of elevation to its top is $28°$. Point B is 30 m closer to the tower on the same straight line, and its angle of elevation is $43°$.

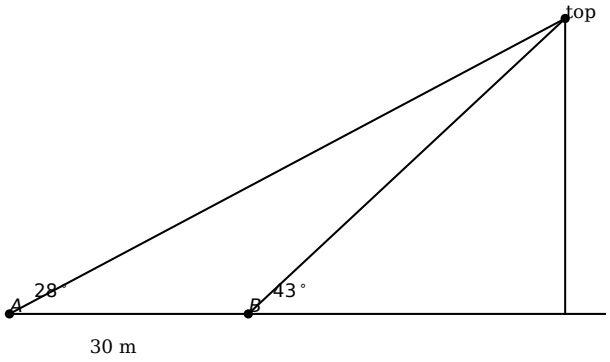


Figure 6

(a) Form an equation in d, the horizontal distance from B to the foot of the tower. (2)

(b) Find the height of the tower to 3 significant figures. (3)

(Total for Question 6 is 5 marks)

7.

In a triangle, two sides have lengths 5 cm and 8 cm and the included angle is θ. Its area is $10\sqrt{3}\ \text{cm}^2$.

(a) Find all possible values of θ in $0^\circ < \theta < 180^\circ$. (3)

(b) Given that the third side is longer than 8 cm, find its exact length. (3)

(Total for Question 7 is 6 marks)

8.

The graph $y = 3\sin x$ is considered for $0^\circ \leq x \leq 360^\circ$.

(a) Sketch the graph, marking all intercepts and all maximum and minimum points. (3)

(b) Find the values of x for which $y = \frac{3}{2}$. (2)

(Total for Question 8 is 5 marks)

A triangular support has sides $AB = AC = 13$ cm and base $BC = 10$ cm.

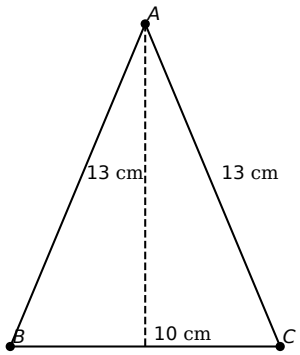


Figure 9

(a) Find the perpendicular height from A to BC and hence the area. (3)

(b) Find $\angle BAC$ to 1 decimal place. (3)

(Total for Question 9 is 6 marks)

Point B is 80 m due north of A. Point C is on a bearing of 065° from A and a bearing of 120° from B.

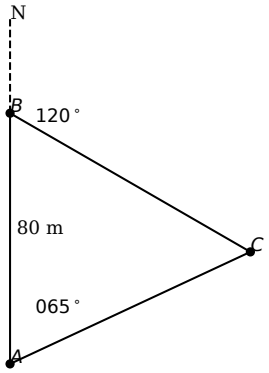


Figure 10

(a) Find all three angles in triangle ABC. (3)

(b) Find AC to 3 significant figures. (2)

(Total for Question 10 is 5 marks)

The curve is $y = 2\cos x - 1$, for $0^\circ \leq x \leq 360^\circ$.

(a) State the maximum and minimum values of y, and all values of x at which they occur. (3)

(b) Find the coordinates where the curve crosses the x-axis. (2)

(Total for Question 11 is 5 marks)

In triangle ABC, $AB = 15$ cm, $AC = 10$ cm and $\angle ABC = 30°$.

(a) Find the two possible values of $\angle ACB$. (3)

(b) For the triangle in which $\angle ACB$ is acute, find

(i) $\angle BAC$ to 1 decimal place, (2)

(ii) the area to 3 significant figures. (2)

(Total for Question 12 is 7 marks)

A ladder of length 7 m leans against a vertical wall on horizontal ground. Its angle with the ground is 64°.

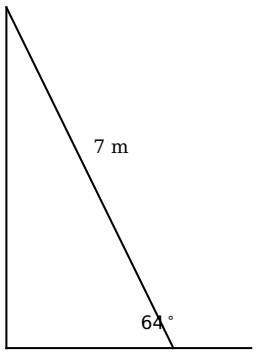


Figure 13

(a) Find the height reached on the wall to 3 significant figures. (2)

(b) The top is lowered by 1 m while the ladder remains against the wall. Find the new angle with the ground to 1 decimal place. (3)

(Total for Question 13 is 5 marks)

A parallelogram has adjacent sides 9 cm and 14 cm and an included acute angle of 38°.

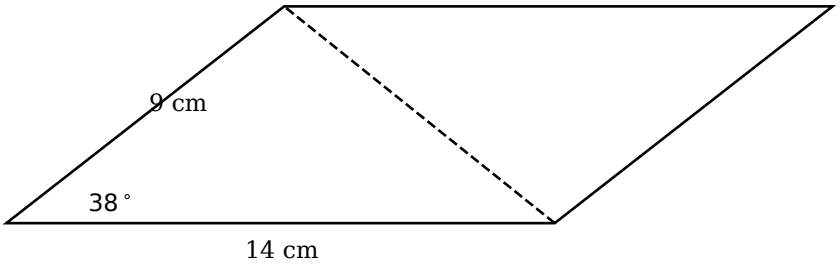


Figure 14

(a) Find its area to 3 significant figures. (2)

(b) Find the length of its shorter diagonal to 3 significant figures. (3)

(Total for Question 14 is 5 marks)

15.

Triangle ABC has $AB = 4$ cm, $AC = 7$ cm and $BC = \sqrt{37}$ cm.

(a) Find the exact value of $\cos A$ and hence find A. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 15 is 5 marks)

16.

For $0^\circ < x < 180^\circ$, $\sin x = \frac{5}{13}$ and x is obtuse.

(a) Find the exact value of $\cos x$. (2)

(b) Find the exact value of $\tan x$. (2)

(Total for Question 16 is 4 marks)

17.

The graph $y = \tan x$ is considered for $-90^\circ < x < 270^\circ$.

(a) Sketch the graph, showing all vertical asymptotes and axis intersections in this interval. (3)

(b) Find all x in the stated interval for which $y = -1$. (2)

(Total for Question 17 is 5 marks)

A triangular field has sides 120 m and 160 m enclosing an angle of $47°$. A straight fence joins the free ends of those sides.

(a) Find the length of the fence to the nearest metre. (3)

(b) A path runs along the 160 m side and has perpendicular width 2 m inside the field. For this calculation only, its area is modelled as a rectangle of dimensions 160 m by 2 m. Find

(i) the modelled path area, (1)

(ii) the percentage of the field area occupied by this modelled path, to 1 decimal place. (3)

(Total for Question 18 is 7 marks)

Two ships leave a harbour at the same time. After 2 hours, ship A has travelled 30 km on a bearing of 020° and ship B has travelled 40 km on a bearing of 110°.

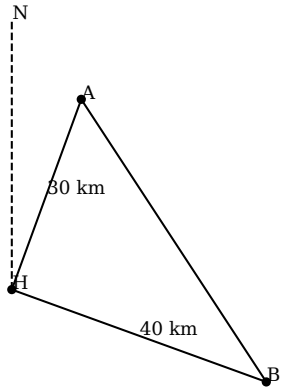


Figure 19

(a) Find their separation at this time. (3)

(b) Find the angle between the direction from A to the harbour and the direction from A to B, to 1 decimal place. (3)

(Total for Question 19 is 6 marks)

A triangle has sides x cm, $2x$ cm and $\sqrt{19}$ cm. The angle between the sides of lengths x and $2x$ is 120°, and $x > 0$.

(a) Find the exact value of x. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 20 is 5 marks)

21. P1-C06-R21 · P1 · Chapter 6

The curve is $y = 4\sin(2x)$, where angles are in degrees and $0^\circ \leq x \leq 180^\circ$.

(a) State its period and the coordinates of its maximum and minimum points on this interval. (3)

(b) Find all zeros on the interval. (2)

(Total for Question 21 is 5 marks)

Triangle ABC has area 24 cm^2, $AB = 8$ cm and $AC = 10$ cm.

(a) Find the two possible values of $\angle BAC$ to 1 decimal place. (3)

(b) Find both possible exact lengths of BC. (3)

(Total for Question 22 is 6 marks)

A mast is held by two wires attached to the same top point. The wires are fixed to the horizontal ground at opposite sides of the mast, at distances 6 m and 10 m from its foot. The shorter wire makes angle 58° with the ground.

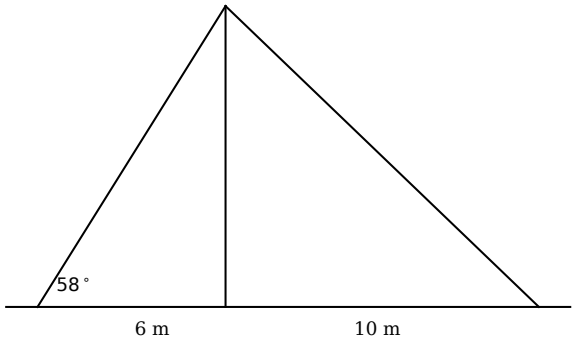


Figure 23

(a) Find the mast height to 3 significant figures. (2)

(b) For the longer wire, find

(i) its angle with the ground to 1 decimal place, (2)

(ii) its length to 3 significant figures. (2)

(Total for Question 23 is 6 marks)

For $0^\circ \leq x \leq 360^\circ$, $\cos x = -\frac{1}{2}$.

(a) Find all possible values of x. (2)

(b) For each value of x found in part (a), state

(i) the exact value of $\sin x$, (2)

(ii) the exact value of $\tan x$. (2)

(Total for Question 24 is 6 marks)

A triangle has sides 7 cm, 8 cm and 9 cm. Let θ be the angle opposite the 8 cm side.

(a) Show that $\cos\theta = \frac{11}{21}$. (2)

(b) Hence find the exact area of the triangle. (3)

(Total for Question 25 is 5 marks)

26.

The height of a point on a wheel is modelled by $h = 5 - 4\cos\theta$ metres, for $0^\circ \leq \theta \leq 360^\circ$.

(a) Find the minimum and maximum heights and state the angle at which each occurs. (3)

(b) Find all angles at which the height is 7 m. (2)

(Total for Question 26 is 5 marks)

$ABCD$ is a quadrilateral with diagonal $AC = 12$ cm. In triangle ABC, $AB = 7$ cm and $BC = 9$ cm. In triangle ACD, $AD = CD = 10$ cm. Points B and D lie on opposite sides of AC.

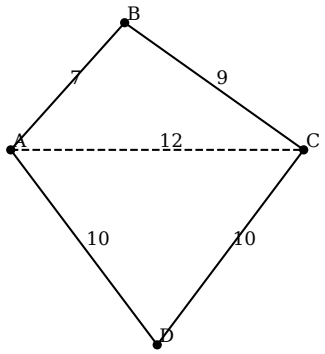


Figure 27

(a) Find $\angle ABC$ to 1 decimal place. (2)

(b) Find the area of the quadrilateral to 3 significant figures. (4)

(Total for Question 27 is 6 marks)

A hillside rises at 18° to the horizontal. A straight path on the hillside is 90 m long.

(a) Find the vertical rise and horizontal distance, each to 3 significant figures. (3)

(b) An alternative route to a platform at the same height consists of a horizontal section of length 100 m followed by a vertical climb. Find its total length to 3 significant figures. (2)

(Total for Question 28 is 5 marks)

Figure 29 shows one complete cycle of $y = a + b\cos x$, for $0^\circ \leq x \leq 360^\circ$, where $b > 0$. The maximum point is $(0, 7)$ and the minimum point is $(180, -1)$.

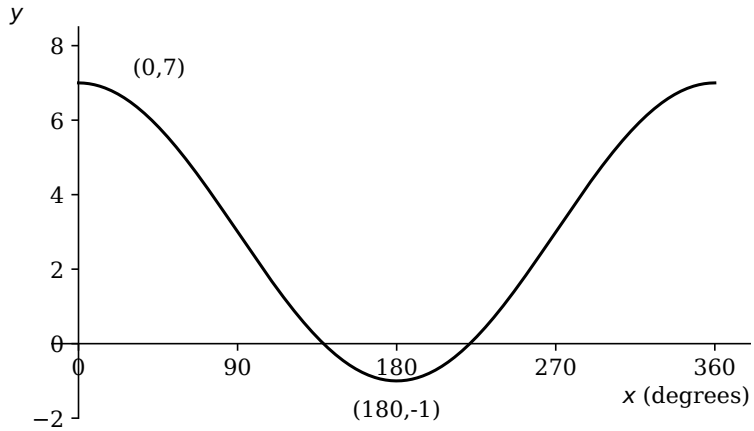


Figure 29

(a) Find a and b. (2)

(b) A horizontal line $y = 5$ cuts this curve at P and Q.

(i) Find the coordinates of P and Q. (2)

(ii) Hence solve $a + b\cos(2t) = 5$ for $0^\circ \leq t \leq 180^\circ$. (1)

(Total for Question 29 is 5 marks)

Triangle ABC has $AB = 8$ cm, $AC = 5$ cm and $\angle BAC = 120°$. Point D lies on BC and AD bisects $\angle BAC$.

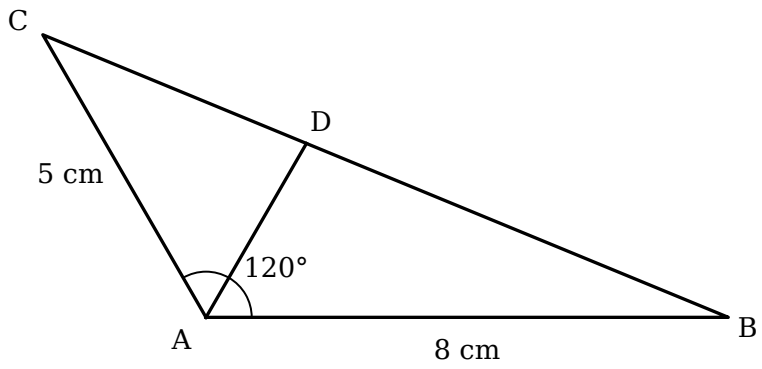


Figure 30

(a) Find the exact length of BC. (2)

(b) Let $AD = x$ cm. Use the areas of triangles ABD and ACD to find

(i) the exact value of x. (3)

(ii) the ratio $BD : DC$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C06-R01 · 5 marks

(a) · 3 marks

$\sqrt{67}$ cm.

余弦定理$BC^2=81+49-2\times9\times7\times1/2=67$。

M1	正确余弦定理
M1	使用60度精确余弦
A1	最简根式

(b) · 2 marks

$\frac{63\sqrt{3}}{4}$ cm^2.

面积$1/2\times9\times7\times\sin60°$，代入$\sqrt{3}/2$。

M1	两边夹角面积式
A1	精确面积

Question 2

P1-C06-R02 · 7 marks

(a) · 2 marks

72°.

三角形内角和180°。

M1	内角和
A1	72度

(b) Giving lengths to 3 significant figures, find

(i) · 2 marks

$12\sin 41^\circ / \sin 67^\circ \approx 8.55$ m.

已知边PQ对67°，PR对41°，正弦定理配对应边角。

M1	正弦定理对应
A1	精度及单位

(ii) · 3 marks

$12 + 12(\sin 41^\circ + \sin 72^\circ)/\sin 67^\circ \approx 33.0$ m.

另边QR对72°，用未舍入的三边相加。

M1	正确求第三边
M1	用未舍入值求周长
A1	周长

Question 3

P1-C06-R03 · 6 marks

(a) · 3 marks

45.8° and 134.2°.

sinC=10sin35°/8≈0.7169705，两角为arcsin及180°−arcsin，且与35°相加均小于180°。

M1	正确正弦关系
M1	找补角
A1	两角及合法检查

(b) · 3 marks

$AC \approx 2.61$ cm.

用未舍入的钝角C得B=180−35−C，再由AC=8sinB/sin35°。

M1	选钝角并求余角
M1	再用正弦定理
A1	长度

Question 4

P1-C06-R04 · 5 marks

(a) · 3 marks

92.1°.

最大角对11；$\cos\theta=(36+81-121)/(2\times6\times9)=-1/27$。

M1	对应最长边与最大角
M1	余弦定理
A1	角度

(b) · 2 marks

$27.0\ \text{cm}^2$.

面积$(1/2)\times6\times9\times\sin\theta$，使用未舍入$\theta$。

M1	正确两边夹角面积
A1	精度及单位

Question 5

P1-C06-R05 · 6 marks

(a) · 2 marks

50°.

正东方位角90°，与040°的夹角50°。

M1	从北向顺时针理解方位角
A1	夹角

(b) · 4 marks

$AB \approx 117$ m; area ≈ 6890 m^2.

用余弦定理求两端距离；面积$1/2 \times 120 \times 150 \times \sin 50^\circ$。

M1	余弦定理
A1	距离
M1	面积式
A1	面积

Question 6

P1-C06-R06 · 5 marks

(a) · 2 marks

$d\tan 43^\circ = (d+30)\tan 28^\circ$.

同一塔高分别为d tan43°与(d+30)tan28°；测量点与塔脚共线，地面水平。

M1	写两种塔高
A1	距离差使用正确

(b) · 3 marks

$h = 30\tan 28^\circ \tan 43^\circ / (\tan 43^\circ - \tan 28^\circ) \approx 37.1$ m.

先解d再乘tan43°，保留中间精度。

M1	解水平距离
M1	求塔高
A1	精度与单位

Question 7

P1-C06-R07 · 6 marks

(a) · 3 marks

$60^\circ, 120^\circ$.

面积$20\sin\theta = 10\sqrt{3}$，所以$\sin\theta = \sqrt{3}/2$，在三角形范围有两个角。

M1	面积方程
M1	两个正弦解
A1	完整角度

(b) · 3 marks

$\sqrt{129}$ cm.

$\theta = 60^\circ$时第三边$^2 = 49$，$\theta = 120^\circ$时为129；长度大于8只允许后者。

M1	分别检验两夹角或比较余弦
M1	用长度条件选支
A1	根式长度

Question 8

P1-C06-R08 · 5 marks

(a) · 3 marks

Intercepts $(0,0),(180,0),(360,0)$; maximum $(90,3)$; minimum $(270,-3)$.

正弦一个完整周期，幅值3，顺序从零上升。

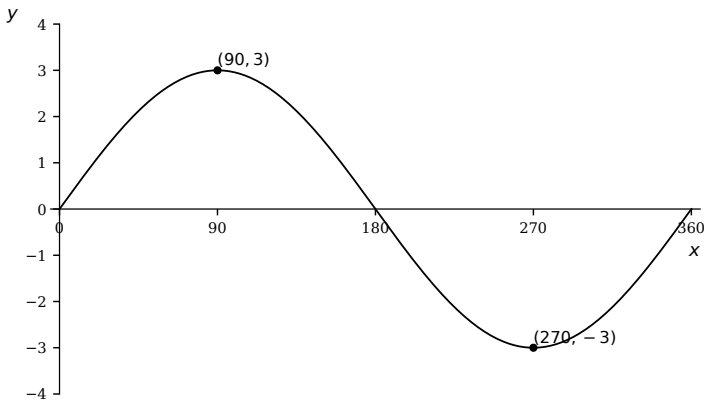
B1	三个零点
B1	两极值
A1	正确正弦形状

(b) · 2 marks

$x = 30^\circ, 150^\circ$.

sinx=1/2，第一和第二象限各一解。

M1	正确化简方程
A1	全部解



Suggested sketch

Question 9

P1-C06-R09 · 6 marks

(a) · 3 marks

Height 12 cm; area 60 cm^2.

等腰三角形高平分底边，半底5，勾股 $\sqrt{(169-25)}=12$。

M1	平分底边并勾股
A1	高
A1	面积

(b) · 3 marks

45.2°.

半顶角满足$\sin(\theta/2)=5/13$，求arcsin再乘2。

M1	正确半角直角三角关系
M1	恢复完整顶角
A1	角度

Question 10

P1-C06-R10 · 5 marks

(a) · 3 marks

$A = 65^\circ$, $B = 60^\circ$, $C = 55^\circ$.

BA从B看向正南180°，与BC的120°夹60°；最后由角和求55°。

M1	正确两处方位夹角
A1	两已知角
A1	第三角

(b) · 2 marks

$80 \sin 60^\circ / \sin 55^\circ \approx 84.6$ m.

AC对B的60°，AB对C的55°，用正弦定理。

M1	正确对应边角
A1	长度

Question 11

P1-C06-R11 · 5 marks

(a) · 3 marks

Maximum 1 at $0^\circ, 360^\circ$; minimum -3 at 180°.

cos取1时最大，取−1时最小，注意区间包含两个端点。

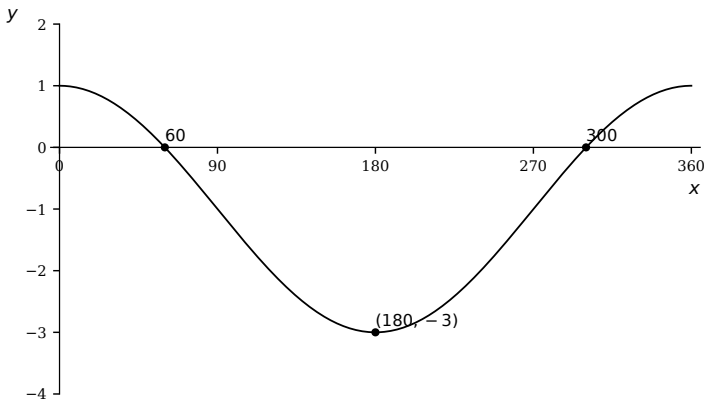
M1	用余弦范围
A1	两极值
A1	所有位置

(b) · 2 marks

$(60, 0)$ and $(300, 0)$, with x in degrees.

令cosx=1/2，第一与第四象限两个解。

M1	化简方程
A1	两个交点



Suggested sketch

Question 12

P1-C06-R12 · 7 marks

(a) · 3 marks

$\sin C = \frac{3}{4}$, so $C \approx 48.6^\circ$ or 131.4°.

正弦定理sinC/15=sin30°/10；两角均与B组成合法三角形。

M1	正弦关系
M1	补角支
A1	两角

(b) For the triangle in which $\angle ACB$ is acute, find

(i) · 2 marks

101.4°.

A=180−30−arcsin(3/4)，用未舍入角。

M1	选正确C并用内角和
A1	A角

(ii) · 2 marks

$\approx 73.5\ \text{cm}^2$.

面积1/2×15×10×sinA。

M1	正确面积式
A1	面积

Question 13

P1-C06-R13 · 5 marks

(a) · 2 marks

$7\sin 64^\circ \approx 6.29$ m.

高度是该角对边，梯长是斜边。

M1	正弦关系
A1	高度

(b) · 3 marks

$\arcsin\big((7\sin 64^\circ - 1)/7\big) \approx 49.1^\circ$.

先从原高度减1，再与不变梯长7构成正弦比；取0°到90°的角。

M1	正确新高度
M1	反求锐角
A1	角度

Question 14

P1-C06-R14 · 5 marks

(a) · 2 marks

$126\sin 38^\circ \approx 77.6\ \text{cm}^2$.

以一边为底、另一边的垂直分量为高；平行四边形面积不是三角形的一半。

M1	正确底高关系
A1	面积

(b) · 3 marks

$\sqrt{9^2+14^2-2(9)(14)\cos 38^\circ} \approx 8.86\ \text{cm}$.

较短对角线对锐角38°，使用余弦定理；另一条对补角更长。

M1	选择锐角对应短对角线
M1	余弦定理
A1	长度

Question 15

P1-C06-R15 · 5 marks

(a) · 3 marks

$\cos A = \frac{1}{2}$, $A = 60^\circ$.

余弦定理(16+49−37)/(2×4×7)=28/56。

M1	正确余弦关系
A1	精确余弦
A1	角度

(b) · 2 marks

$7\sqrt{3}\ \text{cm}^2$.

面积1/2×4×7×√3/2。

M1	正确面积式
A1	根式面积

Question 16

P1-C06-R16 · 4 marks

(a) · 2 marks

$-\frac{12}{13}$.

参考直角三角形对边5斜边13，邻边12；钝角的余弦为负。

M1	勾股得到参考边12
A1	正确负号与比值

(b) · 2 marks

$-\frac{5}{12}$.

tan为sin/cos，正除负得负。

M1	正确三角比商
A1	最简值

Question 17

P1-C06-R17 · 5 marks

(a) · 3 marks

Asymptotes $x = -90^\circ, 90^\circ, 270^\circ$ at boundaries or inside; zeros $0^\circ, 180^\circ$.

正切周期180°，奇数90°为竖直渐近线，各分支严格上升；边界虽不取值仍标渐近线。

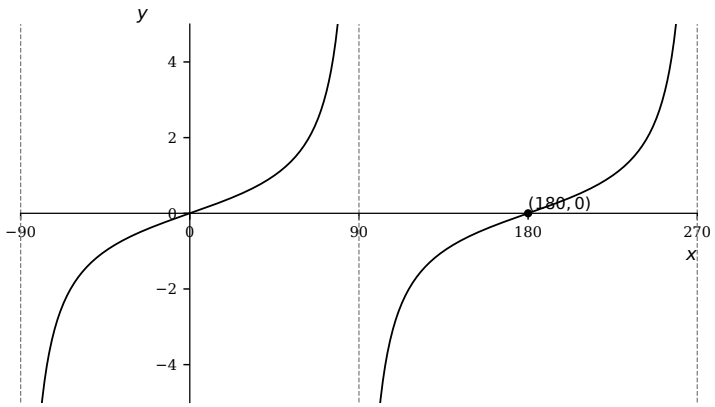
B1	渐近线
B1	两零点
A1	正确分支走向

(b) · 2 marks

$x = -45^\circ, 135^\circ$.

tan为−1的参考角45°，每180°重复，筛选开区间。

M1	使用周期并筛选
A1	全部解



Suggested sketch

Question 18

P1-C06-R18 · 7 marks

(a) · 3 marks

118 m.

余弦定理给 $\sqrt{(120^2+160^2-2\times120\times160\cos47^\circ)}$，再按米四舍五入。

M1	正确余弦式
M1	计算未舍入长度
A1	最近整数米

(b) A path runs along the 160 m side and has perpendicular width 2 m inside the field. For this calculation only, its area is modelled as a rectangle of dimensions 160 m by 2 m. Find

(i) · 1 marks

$320\ \text{m}^2$.

按明确给出的矩形近似模型计算。

B1	矩形面积

(ii) · 3 marks

$100(320)/(9600\sin47^\circ)\approx4.6\%$.

全场面积为$9600\sin47^\circ$；用近似路径面积除以全场面积再乘100。

M1	正确三角场地面积
M1	百分比关系
A1	百分比

Question 19

P1-C06-R19 · 6 marks

(a) · 3 marks

50 km.

夹角110−20=90°，勾股 $\sqrt{(30^2+40^2)}=50$。

M1	从方位得直角
M1	勾股
A1	距离

(b) · 3 marks

53.1°.

在A处夹角的对边HB=40，斜边AB=50，所以sin角=4/5；取三角形锐角。

M1	正确识别A处角的边
M1	反求角
A1	精度

Question 20

P1-C06-R20 · 5 marks

(a) · 3 marks

$x = \sqrt{19/7}$.

余弦定理$19=x^2+4x^2-4x^2\cos120^\circ=7x^2$，取正根。

M1	正确含参数余弦式
M1	使用精确余弦
A1	正根

(b) · 2 marks

$\dfrac{19\sqrt{3}}{14}\ \text{cm}^2$.

面积$x^2\sin120^\circ=(19/7)(\sqrt{3}/2)$。

M1	正确面积式
A1	精确结果

Question 21

P1-C06-R21 · 5 marks

(a) · 3 marks

Period $180°$; maximum $(45, 4)$, minimum $(135, -4)$.

输入角2x走完360°时x走180°，故周期180°；极值在2x=90°,270°。

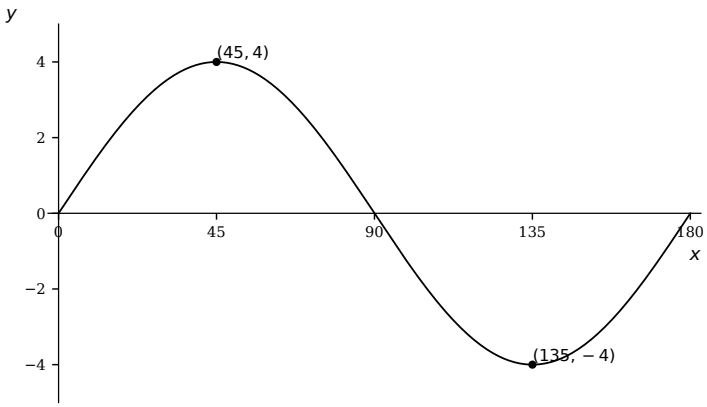
M1	正确处理周期
A1	极值位置
A1	极值坐标

(b) · 2 marks

$x = 0°, 90°, 180°$.

2x=0°,180°,360°，横坐标除2。

M1	输入角范围及零点
A1	全部x



Suggested sketch

Question 22

P1-C06-R22 · 6 marks

(a) · 3 marks

36.9° or 143.1°.

$40\sin A=24$，故$\sin A=3/5$；在0°到180°内有锐角和其补角。

M1	正确面积关系
M1	找两个角
A1	角度

(b) · 3 marks

6 cm and $2\sqrt{73}$ cm.

相应$\cos A=\pm 4/5$。余弦定理$BC^2=164-160\cos A$，分别36和292。

M1	找两余弦的正负
M1	余弦定理
A1	两精确长度

Question 23

P1-C06-R23 · 6 marks

(a) · 2 marks

$6\tan 58^\circ \approx 9.60$ m.

短线对应较近锚点6m；高度为6tan58°。

M1	正确识别短线锚点
A1	高度

(b) For the longer wire, find

(i) · 2 marks

$\arctan\left((6\tan 58^\circ)/10\right) \approx 43.8^\circ$.

同一塔高除以较远水平距离10构成正切比。

M1	正确正切关系
A1	角度

(ii) · 2 marks

$\sqrt{100 + 36\tan^2 58^\circ} \approx 13.9$ m.

水平直角边10与塔高用勾股。

M1	勾股求线长
A1	长度

Question 24

P1-C06-R24 · 6 marks

(a) · 2 marks

$120^\circ, 240^\circ$.

余弦负在第二和第三象限，参考角60°。

M1	正确参考角与象限
A1	全部解

(b) For each value of x found in part (a), state

(i) · 2 marks

$\sqrt{3}/2$ at 120°; $-\sqrt{3}/2$ at 240°.

参考角60°，正弦在第二象限正、第三负。

M1	正确参考值
A1	与两角配对

(ii) · 2 marks

$-\sqrt{3}$ at 120°; $\sqrt{3}$ at 240°.

用$\sin/\cos$或参考角符号规则。

M1	正确比值及符号
A1	两值

Question 25

P1-C06-R25 · 5 marks

(a) · 2 marks

$\cos\theta = (7^2 + 9^2 - 8^2)/(2\cdot 7\cdot 9) = 11/21.$

分子49+81−64=66，分母126，约为11/21。

M1	正确余弦定理
A1	精确化简

(b) · 3 marks

$12\sqrt{5}\ \text{cm}^2.$

θ是三角形内角且cos为正，所以sinθ=√(441−121)/21=8√5/21；面积(1/2)×7×9×该比。

M1	由正弦余弦关系或参考直角三角形得正弦
M1	正确面积式
A1	精确面积

Question 26

P1-C06-R26 · 5 marks

(a) · 3 marks

Minimum 1 m at $0^\circ, 360^\circ$; maximum 9 m at 180°.

余弦前系数负，cos=1时最低，cos=−1时最高；包含两端点。

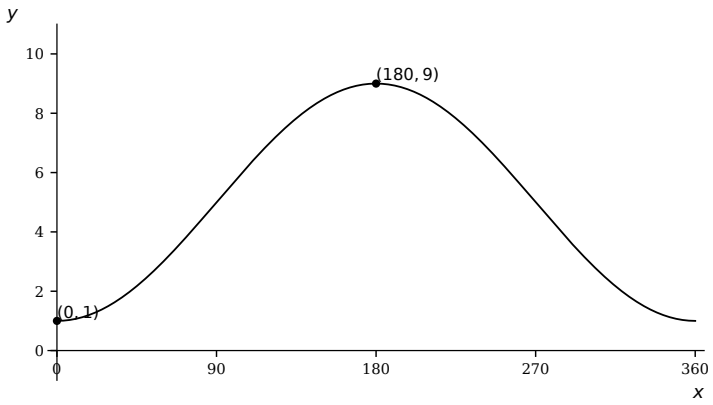
M1	正确使用负系数
A1	两高度
A1	所有角度

(b) · 2 marks

$120^\circ, 240^\circ$.

$5-4\cos\theta=7$给$\cos\theta=-1/2$。

M1	正确化简
A1	两角



Suggested sketch

Question 27

P1-C06-R27 · 6 marks

(a) · 2 marks

96.4°.

AC为角B对边，cosB=(49+81−144)/(2×7×9)=−1/9。

M1	正确余弦式
A1	角度

(b) · 4 marks

$\approx 79.3\ \text{cm}^2$.

ABC面积=(1/2)×7×9sinB=14√5≈31.305；ACD高√(100−36)=8，面积48；相加约79.3。

M1	ABC面积
M1	等腰分割求ACD面积
M1	两三角形相加
A1	总面积

Question 28

P1-C06-R28 · 5 marks

(a) · 3 marks

Rise 27.8 m; horizontal distance 85.6 m.

以路径为斜边，竖直分量$90\sin 18^\circ$，水平$90\cos 18^\circ$。

M1	两分量正确三角比
A1	竖直距离
A1	水平距离

(b) · 2 marks

128 m.

平台与原终点同高，所以竖直爬升仍为$90\sin 18^\circ$；两段总长$100+90\sin 18^\circ \approx 127.812$。

M1	给定段长相加
A1	长度

Question 29

P1-C06-R29 · 5 marks

(a) · 2 marks

$a = 3, b = 4.$

最大、最小值分别为$a + b = 7, a - b = -1$；联立得$a = 3, b = 4$。

M1	两个极值方程
A1	参数

(b) A horizontal line $y = 5$ cuts this curve at P and Q.

(i) · 2 marks

$(60, 5), (300, 5).$

$3 + 4\cos x = 5$，即$\cos x = 1/2$，区间内$x = 60^\circ, 300^\circ$。

M1	化为基本余弦方程
A1	两点

(ii) · 1 marks

$t = 30^\circ, 150^\circ.$

$2t$在原图区间内，对应$2t = 60^\circ, 300^\circ$；除以2。

B1	利用前问并正确变换区间

Question 30

P1-C06-R30 · 7 marks

(a) · 2 marks

$\sqrt{129}$ cm.

余弦定理$BC^2 = 8^2 + 5^2 - 2(8)(5)\cos 120^\circ = 129$。

M1	余弦定理
A1	精确长度

(b) Let $AD = x$ cm. Use the areas of triangles ABD and ACD to find

(i) · 3 marks

$x = 40/13$.

大三角形面积$10\sqrt{3}$。两小三角形面积之和为$\tfrac{1}{2}(8)x\sin 60^\circ + \tfrac{1}{2}(5)x\sin 60^\circ = 13x\sqrt{3}/4$；所以$x = 40/13$。不需要引用角平分线长度公式。

M1	两小面积相加
M1	与大三角形面积相等
A1	x

(ii) · 2 marks

$8:5$.

两小三角形对共同底线BC有同高，面积比等于底边比；其面积由前式给出比$8x:5x = 8:5$。

M1	等高三角形面积比
A1	比值